

Database Management System

EXPERIMENT 1 CREATING AND MANAGING TABLES

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1. Create EMPLOYEES table

```
CREATE TABLE employees (  
    employee_id NUMBER(6) NOT NULL,  
    first_name VARCHAR2(20),  
    last_name VARCHAR2(25) NOT NULL,  
    email VARCHAR2(25) NOT NULL,  
    phone_number VARCHAR2(20),  
    hire_date DATE NOT NULL,  
    job_id VARCHAR2(10) NOT NULL,  
    salary NUMBER(8,2),  
    commission_pct NUMBER(2,2),  
    manager_id NUMBER(6),  
    department_id NUMBER(4)  
);
```

Expected Output:

Table created.

Verify table structure:

```
DESC employees;
```

Expected Output:

Name	Null?	Type
-----	-----	-----
EMPLOYEE_ID	NOT NULL	NUMBER(6)
FIRST_NAME		VARCHAR2(20)
LAST_NAME	NOT NULL	VARCHAR2(25)
EMAIL	NOT NULL	VARCHAR2(25)
PHONE_NUMBER		VARCHAR2(20)
HIRE_DATE	NOT NULL	DATE
JOB_ID	NOT NULL	VARCHAR2(10)
SALARY		NUMBER(8,2)
COMMISSION_PCT		NUMBER(2,2)
MANAGER_ID		NUMBER(6)
DEPARTMENT_ID		NUMBER(4)

2. Create DEPARTMENT table

```
CREATE TABLE department (  
    dept_id NUMBER(6) NOT NULL,  
    dept_name VARCHAR2(20) NOT NULL,  
    manager_id NUMBER(6),  
    location_id NUMBER(4)  
);
```

Expected Output:

Table created.

Verify table structure:

DESC department;

Expected Output:

Name	Null?	Type
DEPT_ID	NOT NULL	NUMBER(6)
DEPT_NAME	NOT NULL	VARCHAR2(20)
MANAGER_ID		NUMBER(6)
LOCATION_ID		NUMBER(4)

3. Create JOB_GRADE table

```
CREATE TABLE job_grade (  
    grade_level VARCHAR2(2),  
    lowest_sal NUMBER,  
    highest_sal NUMBER  
);
```

Expected Output:

Table created.

4. Create LOCATION table

```
CREATE TABLE location (  
    location_id NUMBER(4) NOT NULL,  
    st_addr VARCHAR2(40),  
    postal_code VARCHAR2(12),  
    city VARCHAR2(30) NOT NULL,  
    state_province VARCHAR2(25),  
    country_id CHAR(2)
```

);

Expected Output:

Table created.

Insert Sample Data:

Insert into EMPLOYEES:

```

INSERT INTO employees VALUES (100, 'Steven', 'King',
'SKING', '515.123.4567',
TO_DATE('17-JUN-1987', 'DD-MON-
YYYY'), 'AD_PRES', 24000, NULL, NULL, 90);
INSERT INTO employees VALUES (101, 'Neena', 'Kochhar',
'NKOCHHAR', '515.123.4568',
TO_DATE('21-SEP-1989', 'DD-MON-
YYYY'), 'AD_VP', 17000, NULL, 100, 90);
INSERT INTO employees VALUES (102, 'Lex', 'De Haan',
'LDEHAAN', '515.123.4569',
TO_DATE('13-JAN-1993', 'DD-MON-
YYYY'), 'AD_VP', 17000, NULL, 100, 90);
INSERT INTO employees VALUES (103, 'Alexander', 'Hunold',
'AHUNOLD', '590.423.4567',
TO_DATE('03-JAN-1990', 'DD-MON-
YYYY'), 'IT_PROG', 9000, NULL, 102, 60);
INSERT INTO employees VALUES (104, 'Bruce', 'Ernst',
'BERNST', '590.423.4568',
TO_DATE('21-MAY-1991', 'DD-MON-
YYYY'), 'IT_PROG', 6000, NULL, 103, 60);
INSERT INTO employees VALUES (107, 'Diana', 'Lorentz',
'DLORENTZ', '590.423.4568',
TO DATE('07-FEB-1999', 'DD-MON-

```

```
YYYY'), 'IT_PROG', 4200, NULL, 103, 60);  
COMMIT;
```

Expected Output:

```
1 row created.  
1 row created.  
1 row created.  
1 row created.  
1 row created.  
1 row created.  
Commit complete.
```

Insert into DEPARTMENT:

```
INSERT INTO department VALUES (10, 'Administration', 200,  
1700);  
INSERT INTO department VALUES (20, 'Marketing', 201, 1800);  
INSERT INTO department VALUES (50, 'Shipping', 124, 1500);  
INSERT INTO department VALUES (60, 'IT', 103, 1400);  
INSERT INTO department VALUES (80, 'Sales', 149, 2500);  
INSERT INTO department VALUES (90, 'Executive', 100, 1700);  
INSERT INTO department VALUES (110, 'Accounting', 205,  
1700);  
COMMIT;
```

Expected Output:

```
1 row created.  
1 row created.  
1 row created.  
1 row created.  
1 row created.
```

1 row created.
1 row created.
Commit complete.

Insert into JOB_GRADE:

```
INSERT INTO job_grade VALUES ('A', 1000, 2999);  
INSERT INTO job_grade VALUES ('B', 3000, 5999);  
INSERT INTO job_grade VALUES ('C', 6000, 9999);  
INSERT INTO job_grade VALUES ('D', 10000, 14999);  
INSERT INTO job_grade VALUES ('E', 15000, 24999);  
INSERT INTO job_grade VALUES ('F', 25000, 40000);  
COMMIT;
```

Expected Output:

1 row created.
1 row created.
1 row created.
1 row created.
1 row created.
1 row created.
Commit complete.

Insert into LOCATION:

```
INSERT INTO location VALUES (1400, '2014 Jabberwocky Rd',  
'26192', 'Southlake', 'Texas', 'US');  
INSERT INTO location VALUES (1500, '2011 Interior Blvd',  
'99236', 'South San Francisco', 'California', 'US');  
INSERT INTO location VALUES (1700, '2004 Charade Rd',  
'98199', 'Seattle', 'Washington', 'US');  
INSERT INTO location VALUES (1800, '460 Bloor St', 'M5S
```

```
1XB', 'Toronto', 'Ontario', 'CA');
INSERT INTO location VALUES (2500, 'Magdalen Centre, The
Oxford Science Park', 'OX9 9ZB', 'Oxford', 'Oxford', 'UK');
COMMIT;
```

Expected Output:

```
1 row created.
1 row created.
1 row created.
1 row created.
1 row created.
Commit complete.
```

Verify Data:

Check EMPLOYEES data:

```
SELECT * FROM employees;
```

Expected Output:

EMPLOYEE_ID	FIRST_NAME	LAST_NAME	EMAIL	PHONE_NUMBER	HIRE_DATE	JOB_ID	SALARY	COMMISSION_PCT	MANAGER_ID	DEPARTMENT_ID
100	Steven	King	SKING	515.123.4567	17-JUN-87	AD_PRES	24000		90	
101	Neena	Kochhar	NKOCHHAR	515.123.4568	21-SEP-89					

AD_VP	17000	100	90
102 Lex		De Haan	
LDEHAAN		515.123.4569	13-JAN-93
AD_VP	17000	100	90
103 Alexander		Hunold	
AHUNOLD		590.423.4567	03-JAN-90
IT_PROG	9000	102	60
104 Bruce		Ernst	
BERNST		590.423.4568	21-MAY-91
IT_PROG	6000	103	60
107 Diana		Lorentz	
DLORENTZ		590.423.4568	07-FEB-99
IT_PROG	4200	103	60

Check DEPARTMENT data:

```
SELECT * FROM department;
```

Expected Output:

DEPT_ID	DEPT_NAME	MANAGER_ID	LOCATION_ID
10	Administration	200	1700
20	Marketing	201	1800
50	Shipping	124	1500
60	IT	103	1400
80	Sales	149	2500
90	Executive	100	1700
110	Accounting	205	1700

Check JOB_GRADE data:

```
SELECT * FROM job_grade;
```

Expected Output:

GR	LOWEST_SAL	HIGHEST_SAL
A	1000	2999
B	3000	5999
C	6000	9999
D	10000	14999
E	15000	24999
F	25000	40000

Check LOCATION data:

```
SELECT * FROM location;
```

Expected Output:

LOCATION_ID	ST_ADDR	STATE_PROVINCE
POSTAL_CODE	CITY	
CO		
1400	2014 Jabberwocky Rd	26192
Southlake	Texas	US
1500	2011 Interior Blvd	99236
South San Francisco	California	US
1700	2004 Charade Rd	98199
Seattle	Washington	US
1800	460 Bloor St	M5S 1XB
Toronto	Ontario	CA

2500 Magdalen Centre, The Oxford Science Park OX9 9ZB
Oxford Oxford UK

Exercise Solutions:

1. Create DEPT table based on the table instance chart

```
CREATE TABLE dept (  
    id NUMBER(7),  
    name VARCHAR2(25)  
);
```

Expected Output:

Table created.

Verify:

```
DESC dept;
```

Expected Output:

Name	Null?	Type
----	-----	-----
ID		NUMBER(7)
NAME		VARCHAR2(25)

2. Create EMP table based on the instance chart

```
CREATE TABLE emp (  
    id NUMBER(7),  
    last_name VARCHAR2(25),  
    first_name VARCHAR2(25),
```

```
dept_id NUMBER(7)
);
```

Expected Output:

Table created.

Verify:

```
DESC emp;
```

Expected Output:

Name	Null?	Type
ID		NUMBER(7)
LAST_NAME		VARCHAR2(25)
FIRST_NAME		VARCHAR2(25)
DEPT_ID		NUMBER(7)

3. Modify EMP table to allow longer employee last names

```
ALTER TABLE emp MODIFY last_name VARCHAR2(50);
```

Expected Output:

Table altered.

Verify:

```
DESC emp;
```

Expected Output:

Name	Null?	Type
ID		NUMBER(7)
LAST_NAME		VARCHAR2(50)
FIRST_NAME		VARCHAR2(25)
DEPT_ID		NUMBER(7)

4. Create EMPLOYEES2 table from EMPLOYEES structure

```
CREATE TABLE employees2 AS
SELECT
    employee_id AS id,
    first_name,
    last_name,
    salary,
    department_id AS dept_id
FROM employees
WHERE 1=0;
```

Expected Output:

Table created.

Verify:

```
DESC employees2;
```

Expected Output:

Name	Null?	Type
------	-------	------

ID	NUMBER(6)
FIRST_NAME	VARCHAR2(20)
LAST_NAME	VARCHAR2(25)
SALARY	NUMBER(8,2)
DEPT_ID	NUMBER(4)

5. Drop the EMP table

```
DROP TABLE emp;
```

Expected Output:

Table dropped.

6. Rename EMPLOYEES2 table as EMP

```
RENAME employees2 TO emp;
```

Expected Output:

Table renamed.

Verify:

```
SELECT table_name FROM user_tables WHERE table_name = 'EMP';
```

Expected Output:

```
TABLE_NAME
-----
EMP
```

7. Add comments on DEPT and EMP tables

```
COMMENT ON TABLE dept IS 'This table stores department
information';
COMMENT ON TABLE emp IS 'This table stores employee
details';
```

Expected Output:

Comment created.
Comment created.

Verify comments:

```
SELECT table_name, comments
FROM user_tab_comments
WHERE table_name IN ('DEPT', 'EMP');
```

Expected Output:

TABLE_NAME	COMMENTS
DEPT	This table stores department information
EMP	This table stores employee details

8. Drop the First_name column from EMP table

```
ALTER TABLE emp DROP COLUMN first_name;
```

Expected Output:

Table altered.

Verify:

```
DESC emp;
```

Expected Output:

Name	Null?	Type
ID		NUMBER(6)
LAST_NAME		VARCHAR2(25)
SALARY		NUMBER(8,2)
DEPT_ID		NUMBER(4)

G. Mark DEPT_ID column as UNUSED in EMP table

```
ALTER TABLE emp SET UNUSED (dept_id);
```

Expected Output:

```
Table altered.
```

Verify:

```
DESC emp;
```

Expected Output:

Name	Null?	Type
ID		NUMBER(6)
LAST_NAME		VARCHAR2(25)
SALARY		NUMBER(8,2)

10. Drop all UNUSED columns from EMP table

```
ALTER TABLE emp DROP UNUSED COLUMNS;
```

Expected Output:

Table altered.

Final Table Status:

List all tables created:

```
SELECT table_name FROM user_tables;
```

Expected Output:

```
TABLE_NAME
-----
EMPLOYEES
DEPARTMENT
JOB_GRADE
LOCATION
DEPT
EMP
```